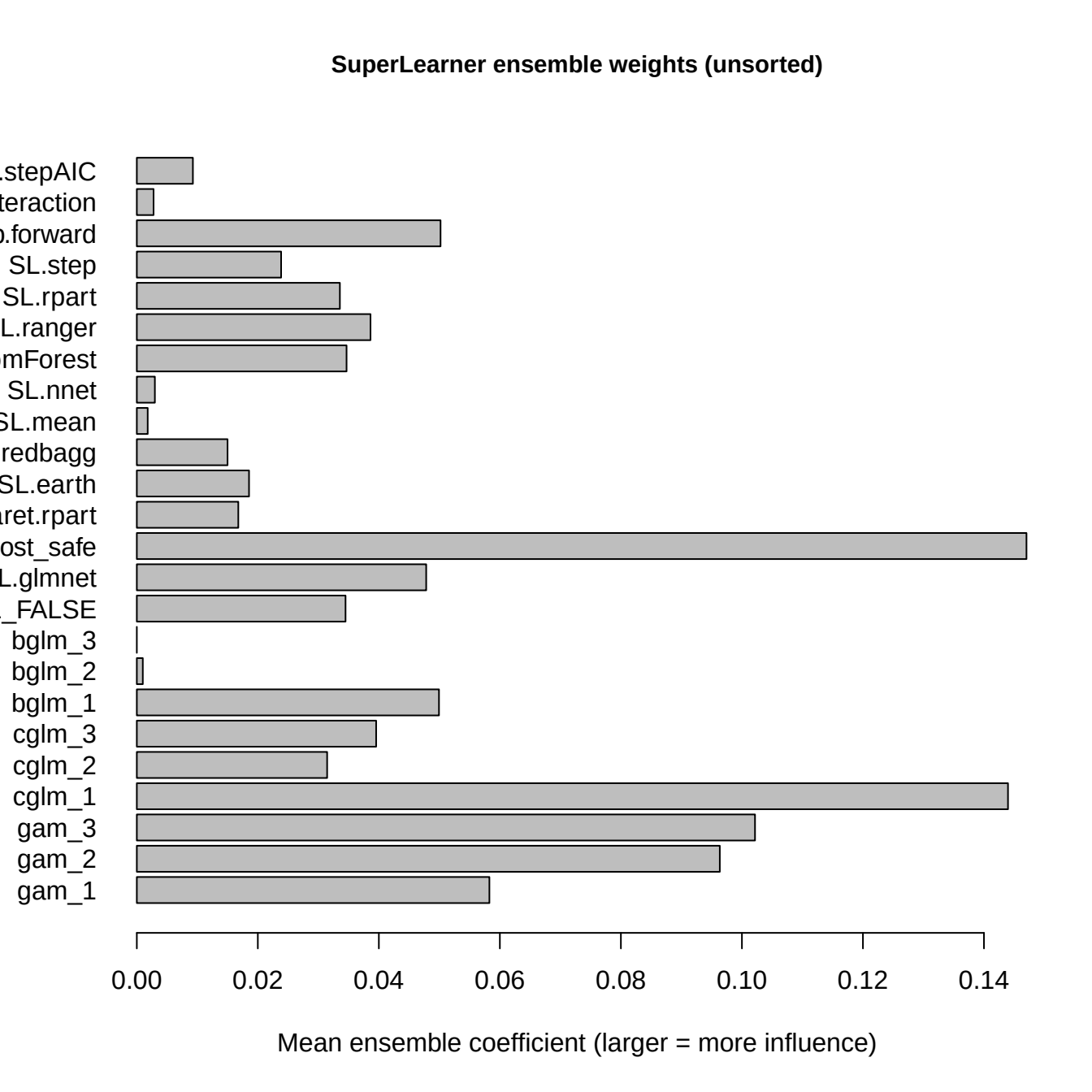
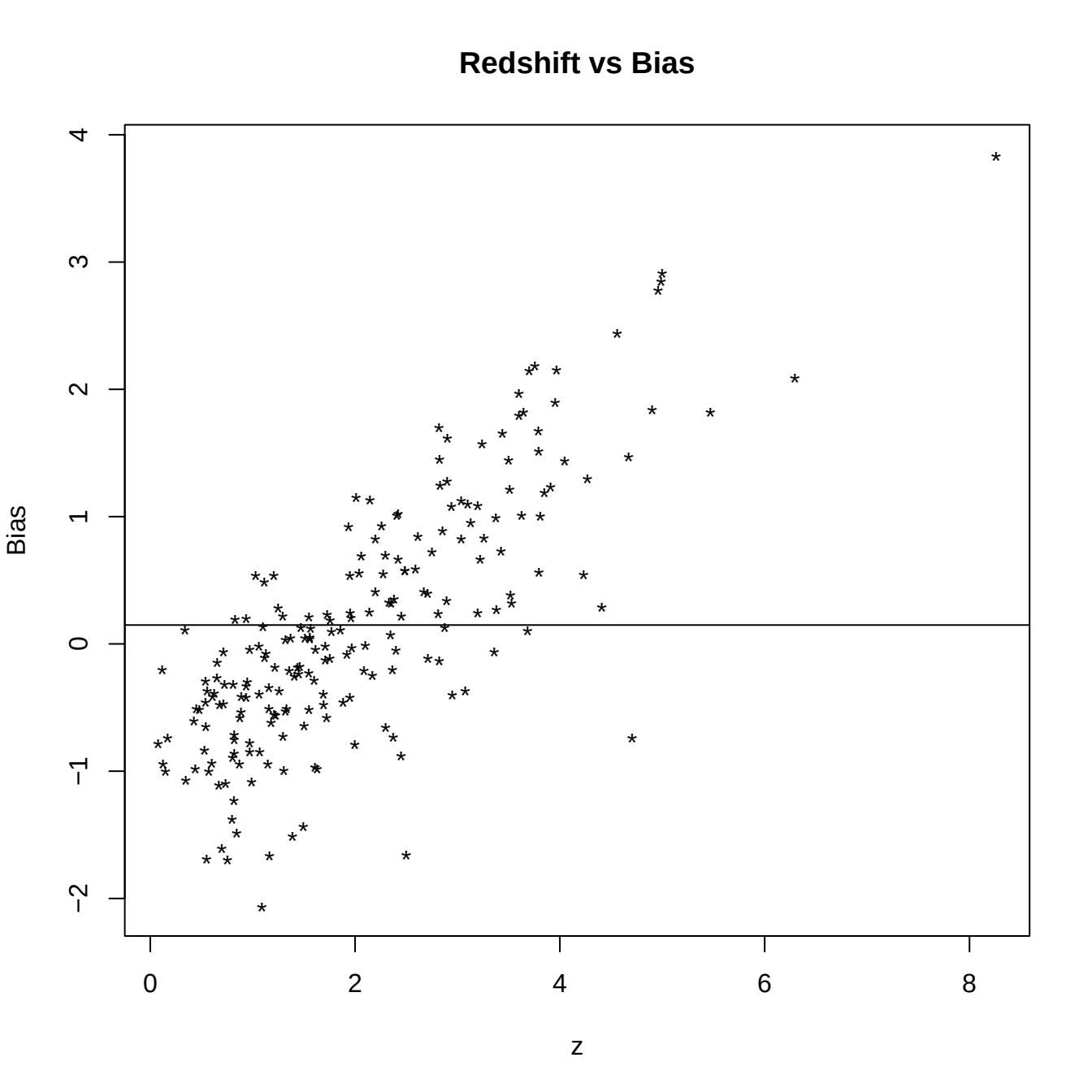
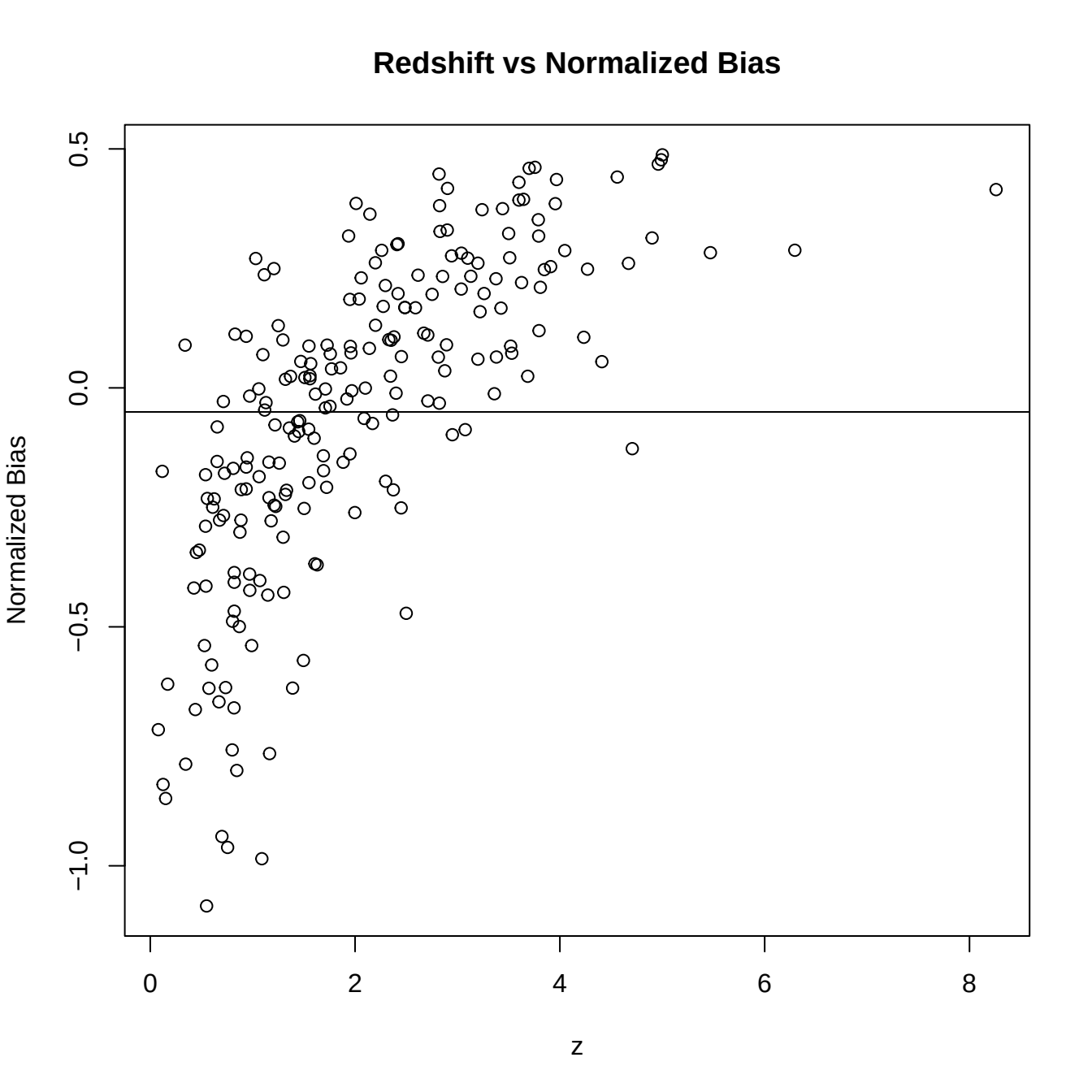
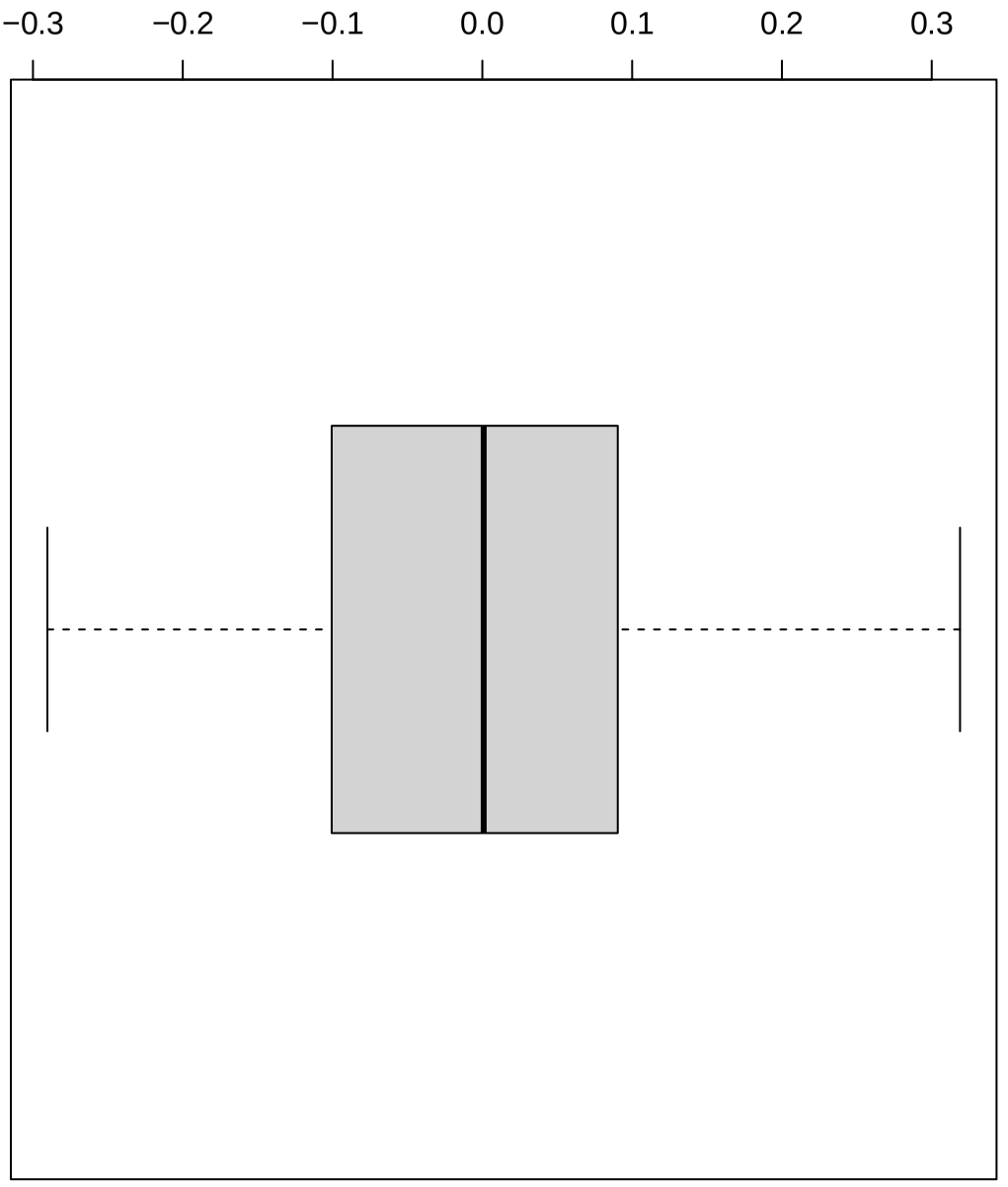
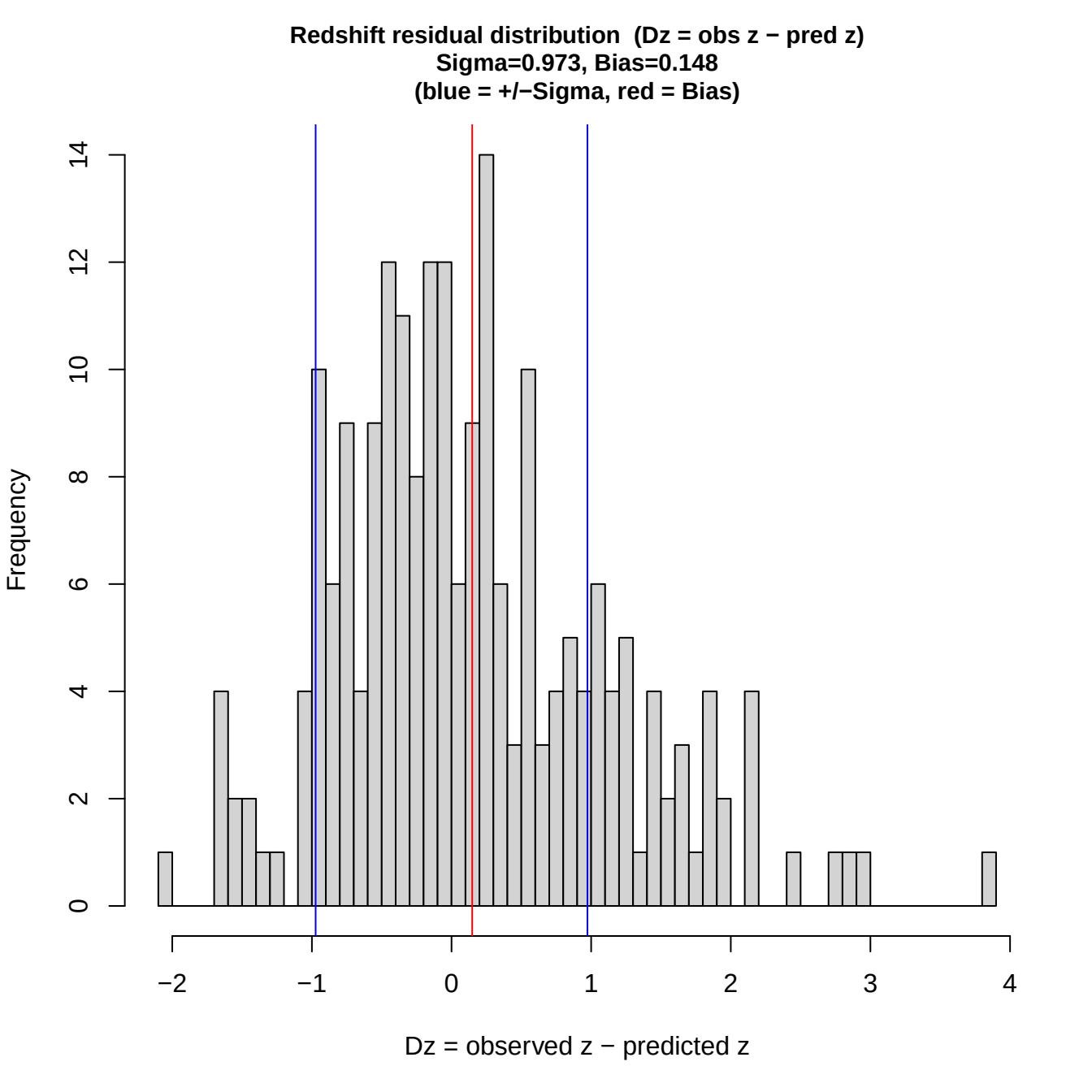
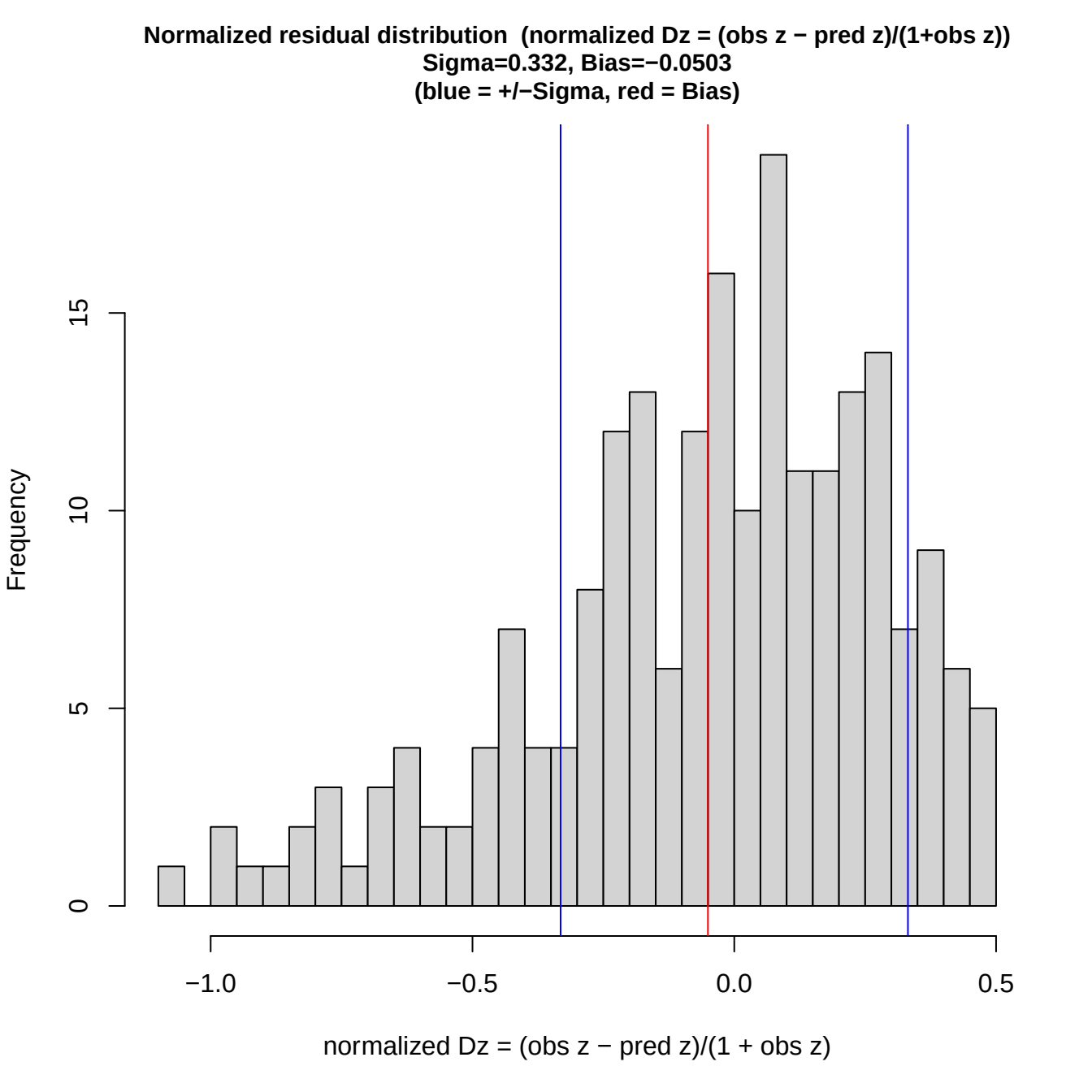

SuperLearner ensemble weights (unsorted)
.stepAIC
teraction
.forward
SL.step
SL.rpart
L.ranger
mForest
SL.nnet
SL.mean
redbagg
SL.earth
aret.rpart
ost_safe
L.glmnet
_FALSE
bglm_3
bglm_2
bglm_1
cglm_3
cglm_2
cglm_1
gam_3
gam_2
gam_1
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Mean ensemble coefficient (larger = more influence)

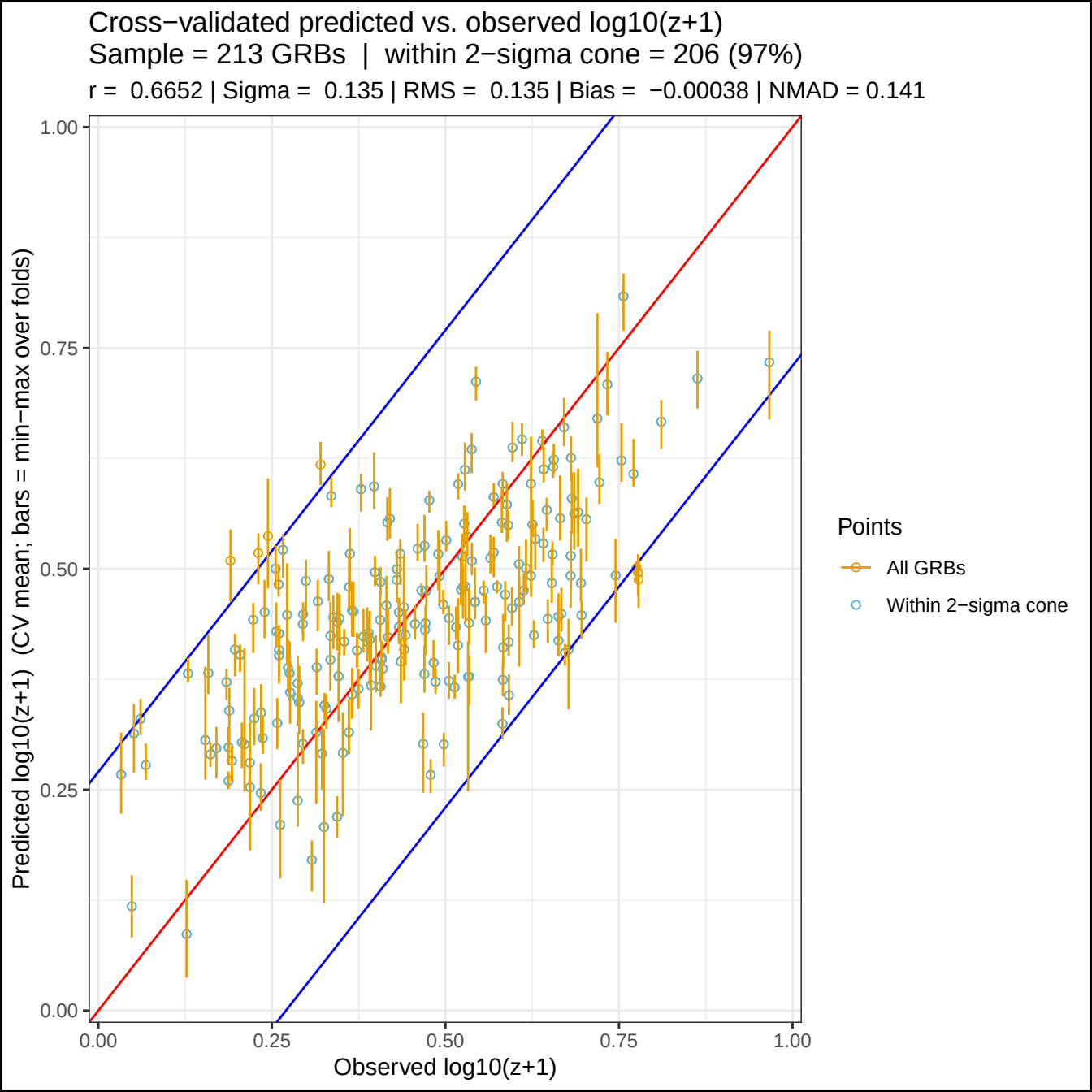

Redshift vs Bias
Bias
z
4
3
2
1
0
−1
−2
0
2
4
6
8

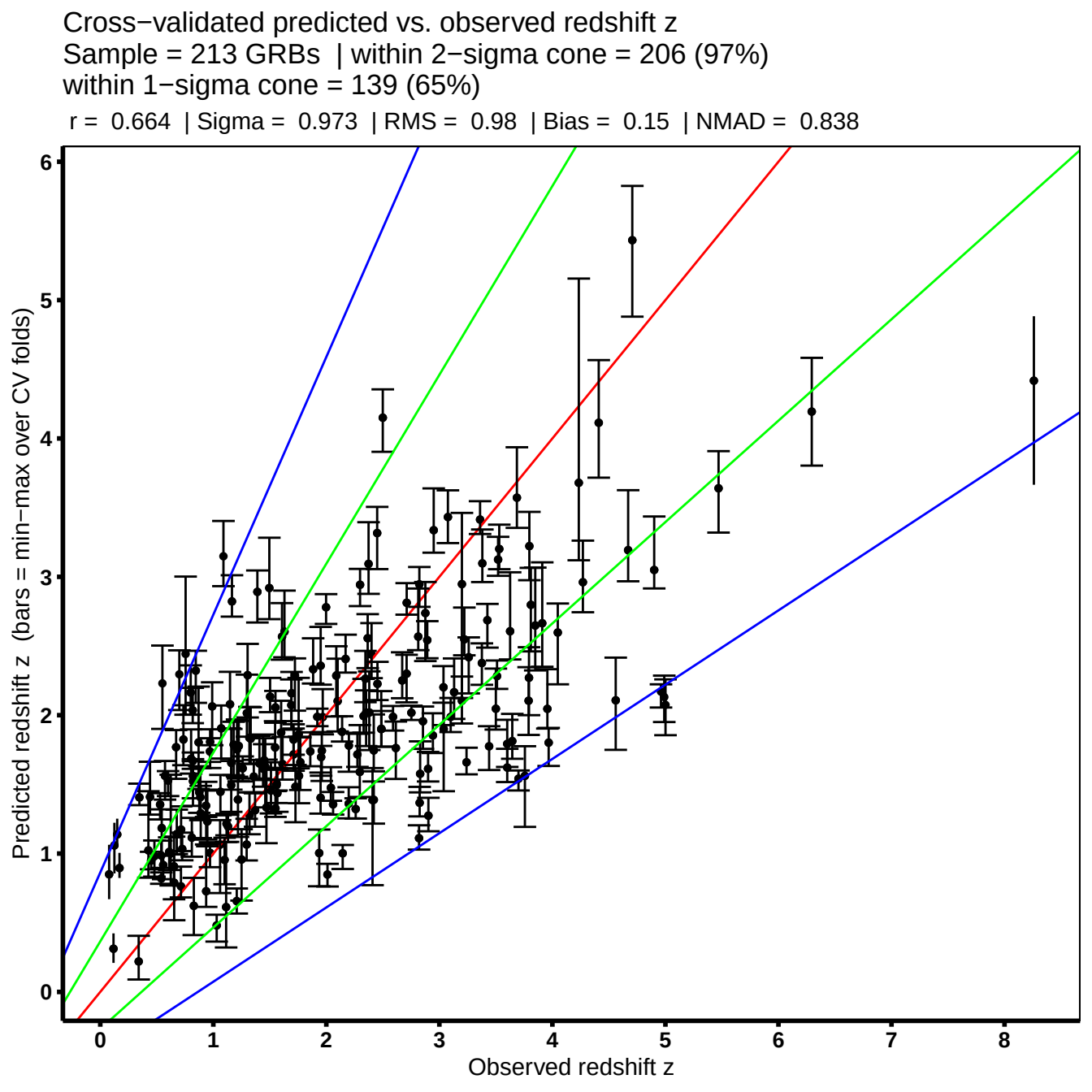

Redshift vs Normalized Bias
Normalized Bias
0.5
0.0
−0.5
−1.0
0
2
4
6
8
z

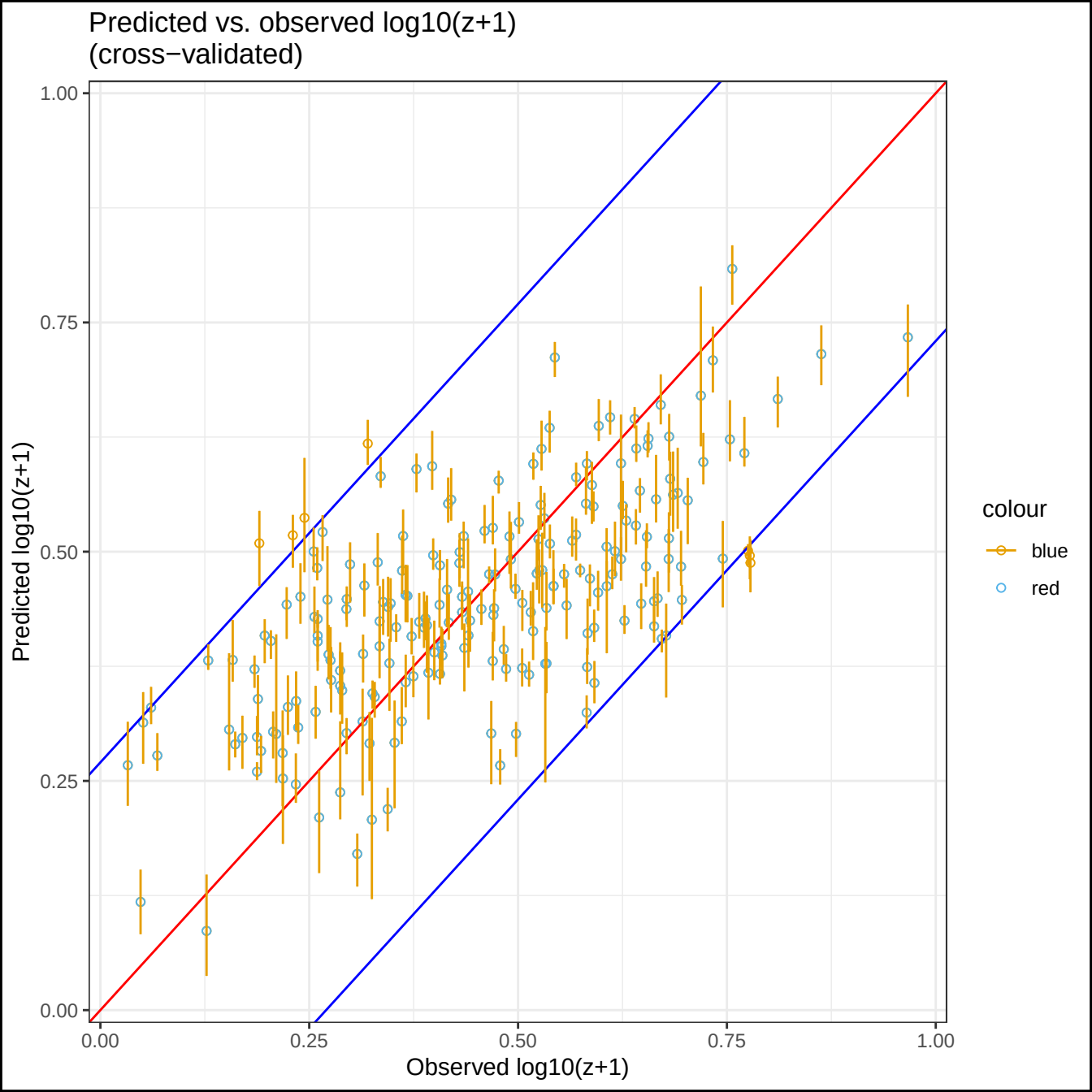

−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3

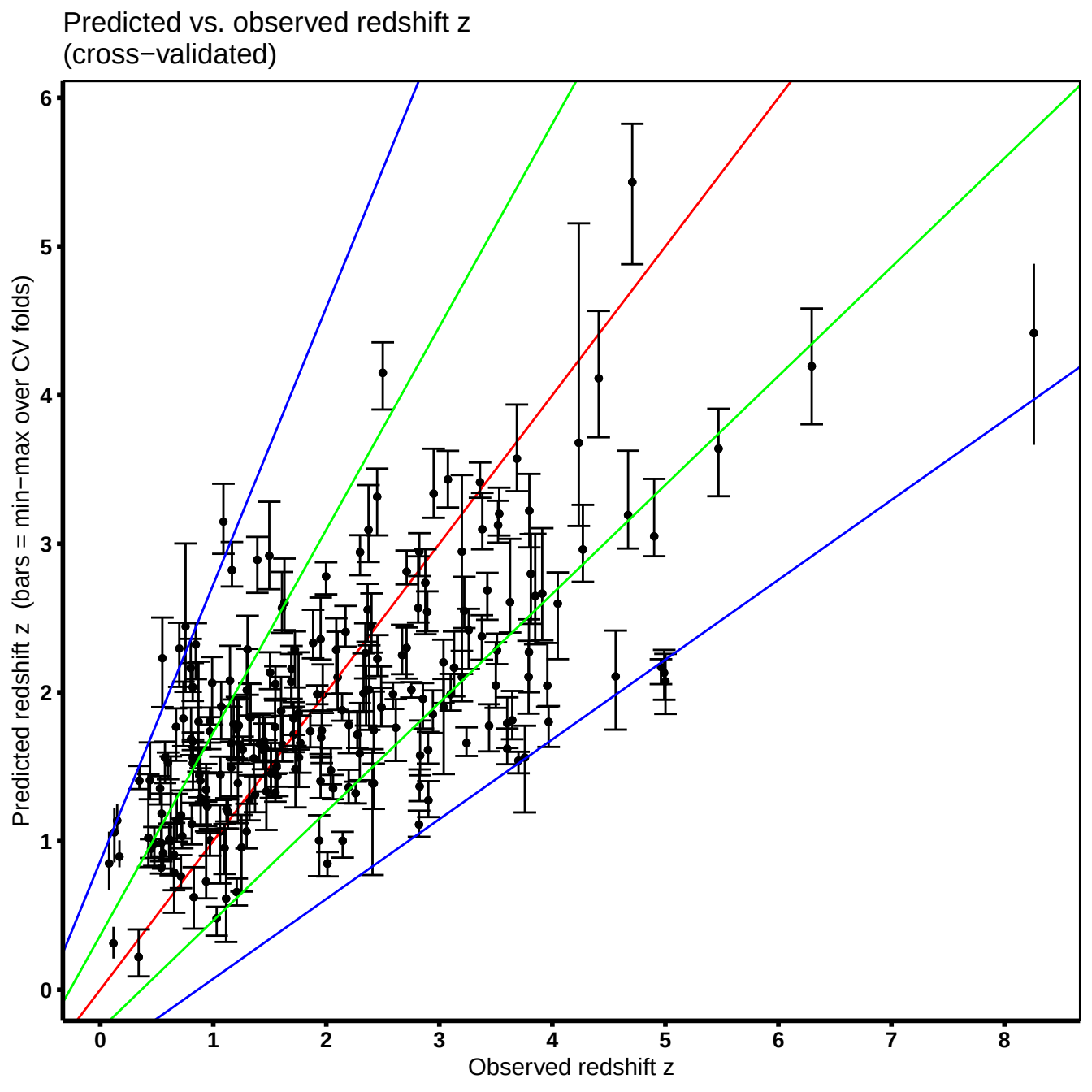

Redshift residual distribution (Dz = obs z − pred z)
Sigma=0.973, Bias=0.148
(blue = +/−Sigma, red = Bias)
Frequency
0
2
4
6
8
10
12
14
−2
−1
0
1
2
3
4
Dz = observed z − predicted z

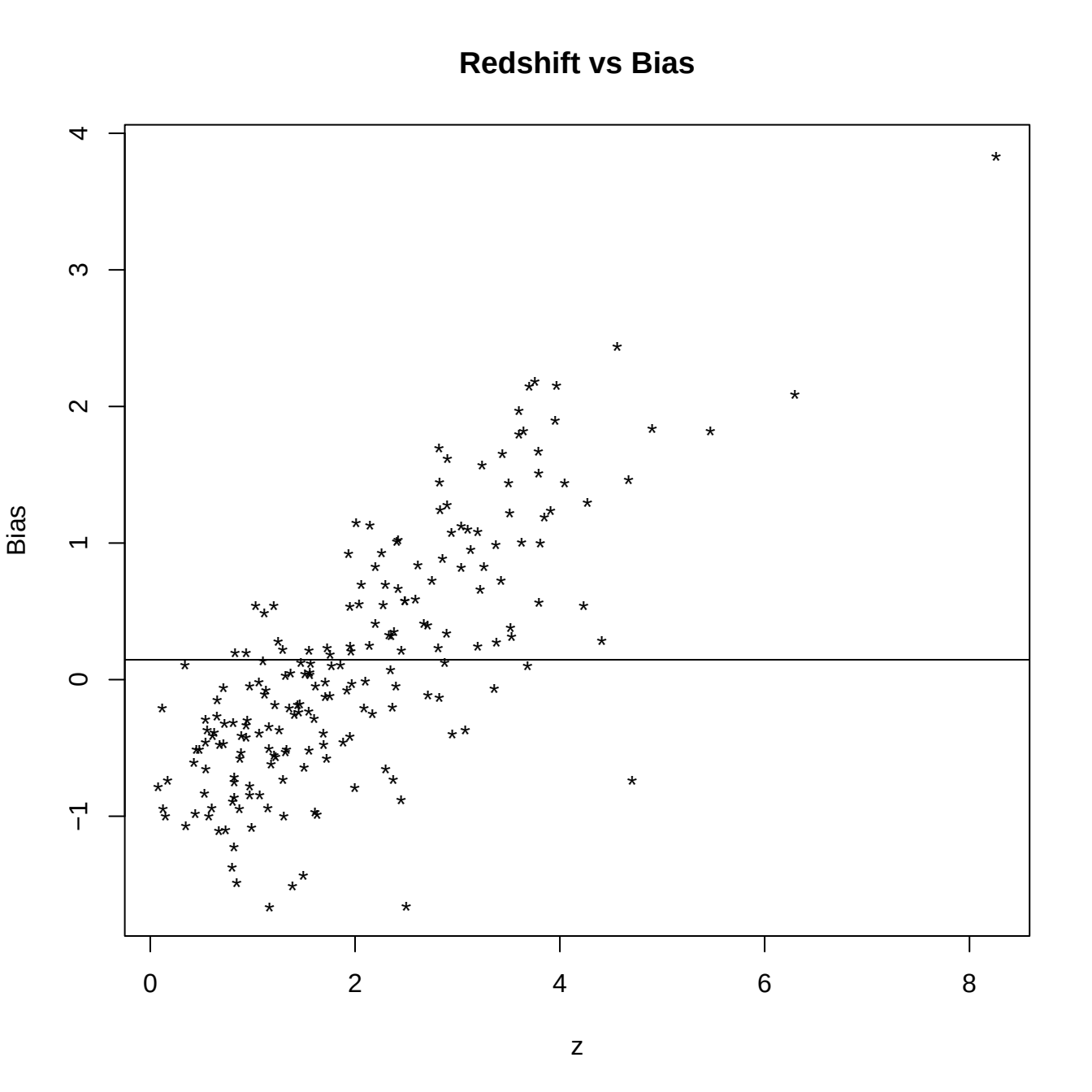

Normalized residual distribution (normalized Dz = (obs z − pred z)/(1+obs z))
Sigma=0.332, Bias=−0.0503
(blue = +/−Sigma, red = Bias)
Frequency
0
5
10
15
−1.0
−0.5
0.0
0.5
normalized Dz = (obs z − pred z)/(1 + obs z)

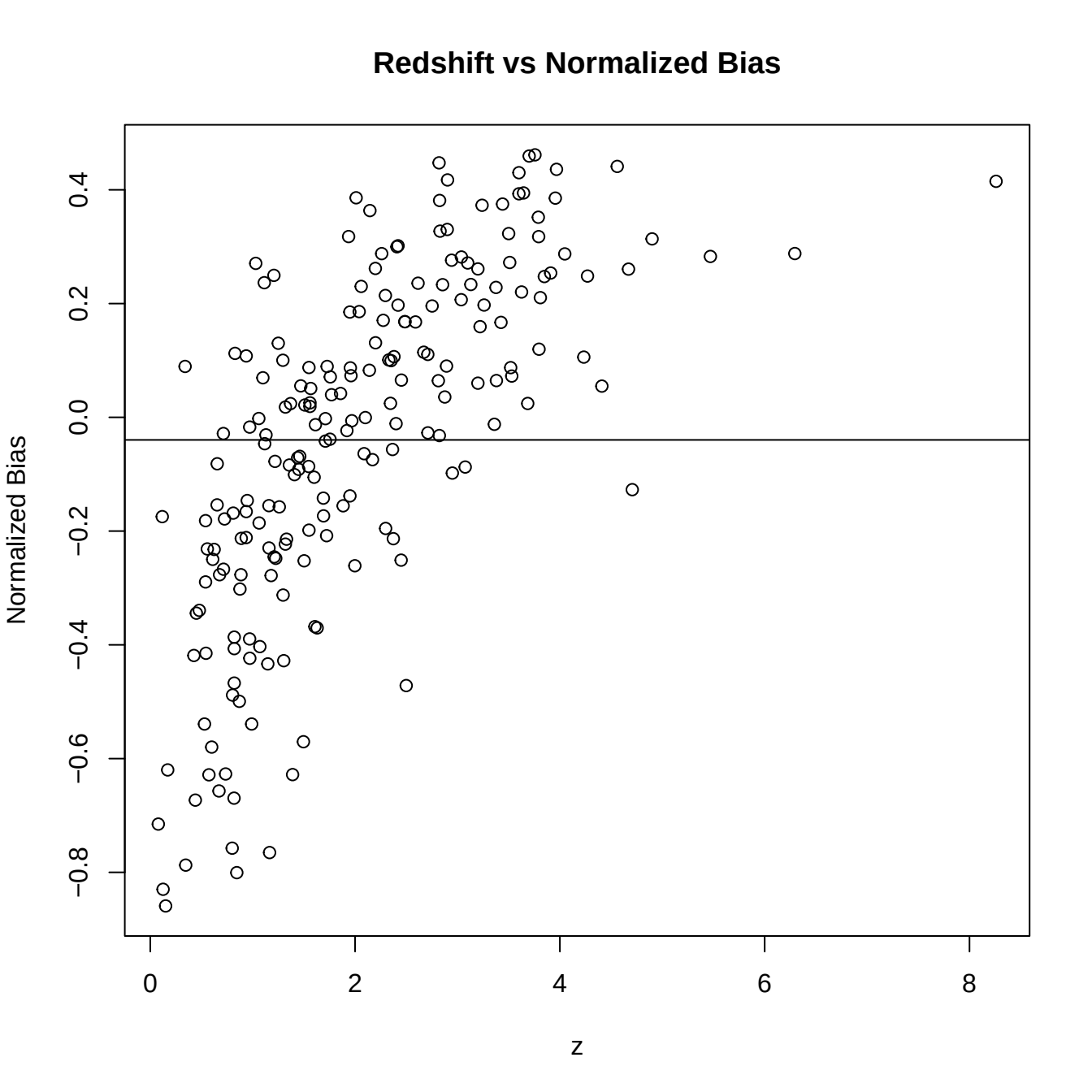

Cross−validated predicted vs. observed log10(z+1)
Sample = 213 GRBs | within 2−sigma cone = 206 (97%)
r = 0.6652 | Sigma = 0.135 | RMS = 0.135 | Bias = −0.00038 | NMAD = 0.141
Predicted log10(z+1) (CV mean; bars = min−max over folds)
Observed log10(z+1)
0.00
0.25
0.50
0.75
1.00
Points
All GRBs
Within 2−sigma cone

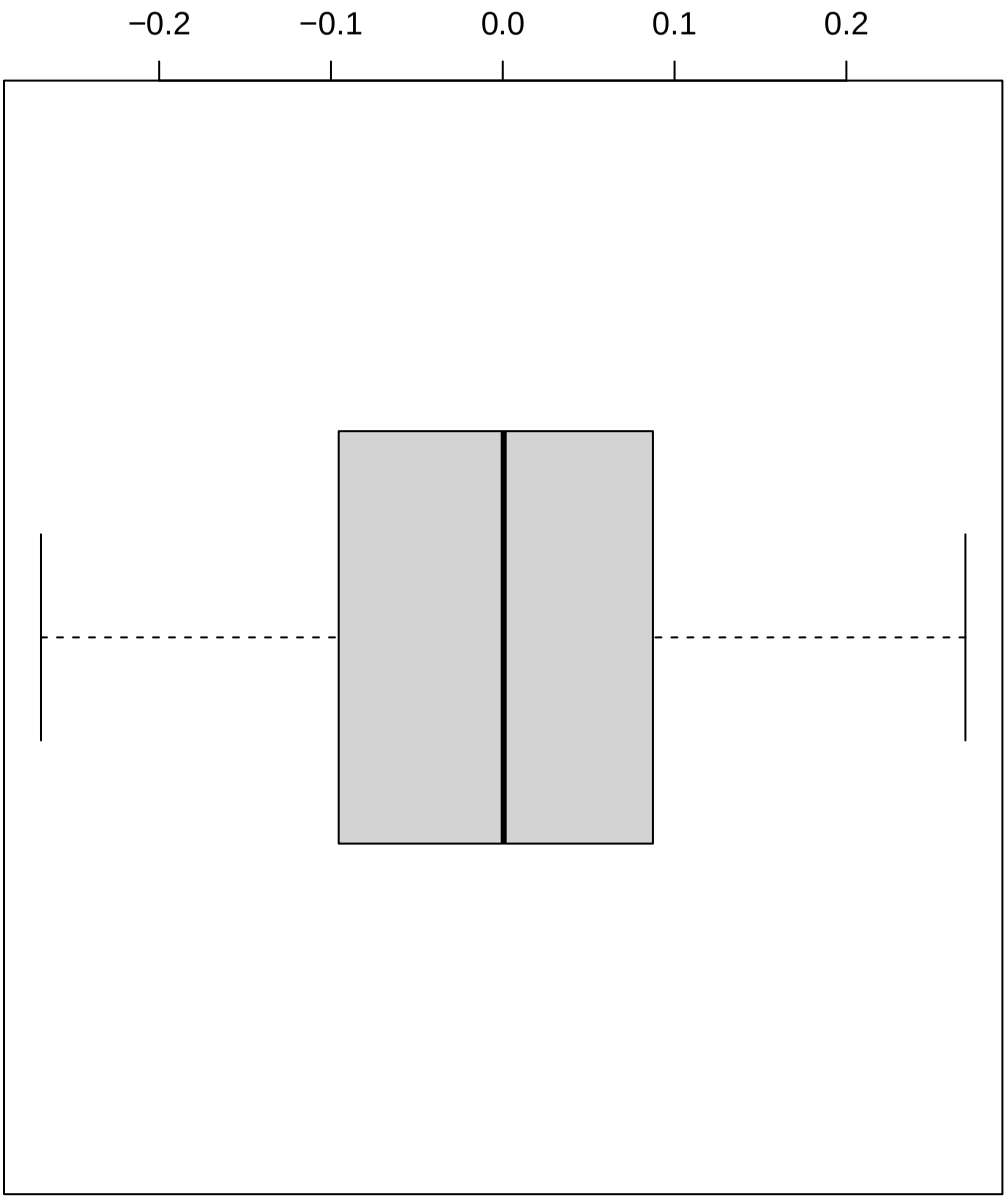

Cross−validated predicted vs. observed redshift z
Sample = 213 GRBs | within 2−sigma cone = 206 (97%)
within 1−sigma cone = 139 (65%)
r = 0.664 | Sigma = 0.973 | RMS = 0.98 | Bias = 0.15 | NMAD = 0.838
Predicted redshift z (bars = min−max over CV folds)
Observed redshift z
0
1
2
3
4
5
6
7
8

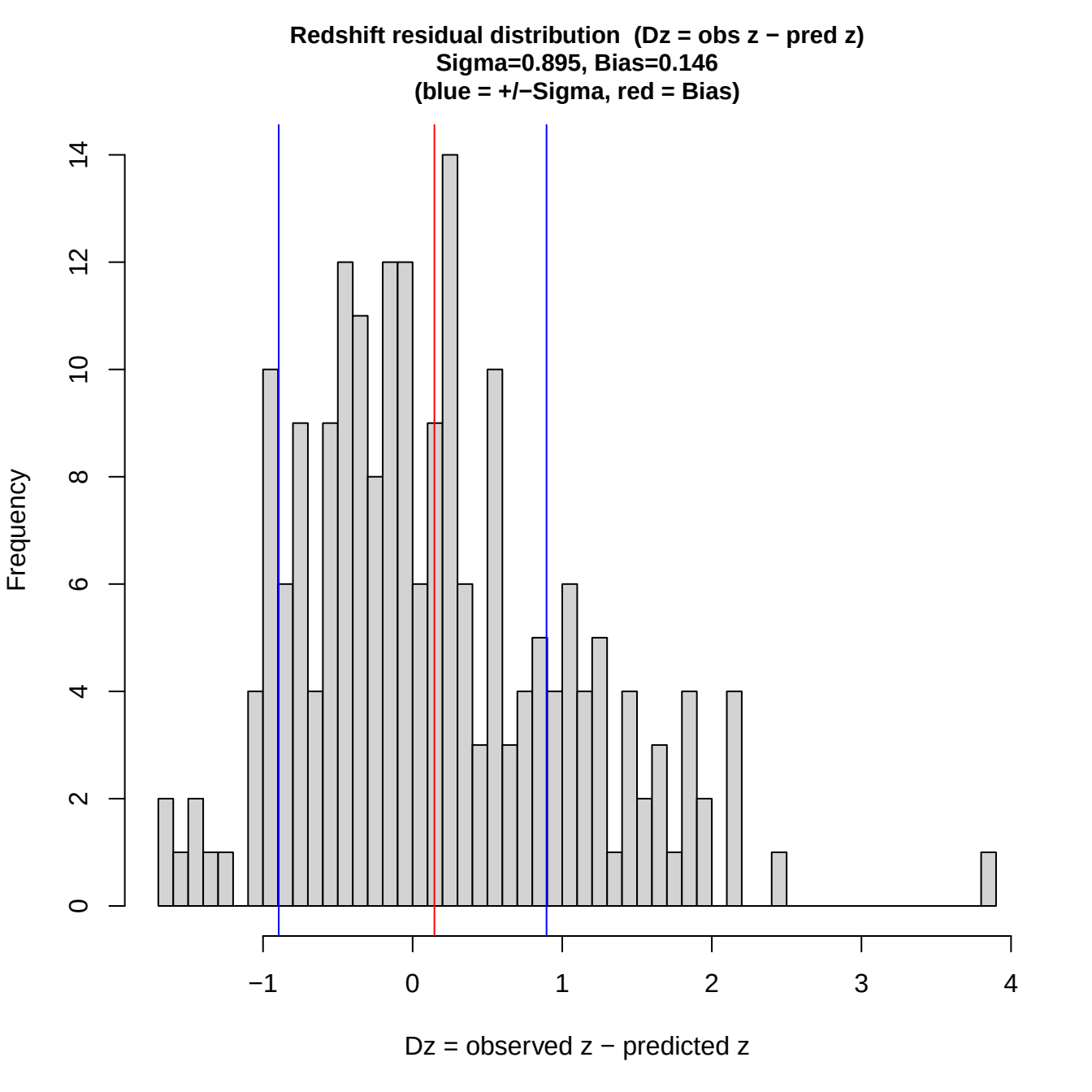

Predicted vs. observed log10(z+1)
(cross−validated)
Predicted log10(z+1)
Observed log10(z+1)
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
colour
blue
red

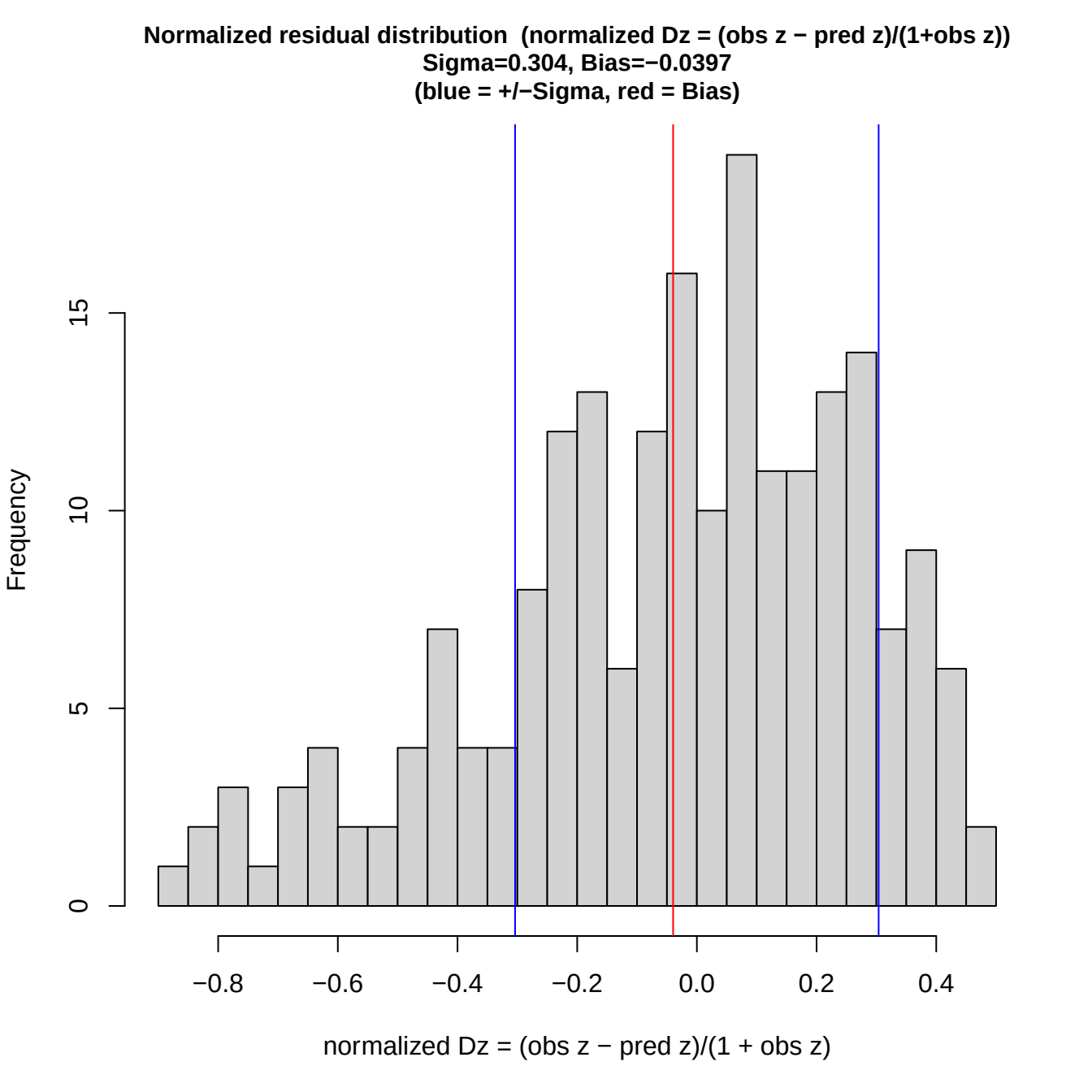

Predicted vs. observed redshift z
(cross−validated)
Predicted redshift z (bars = min−max over CV folds)
Observed redshift z
0
1
2
3
4
5
6
7
8

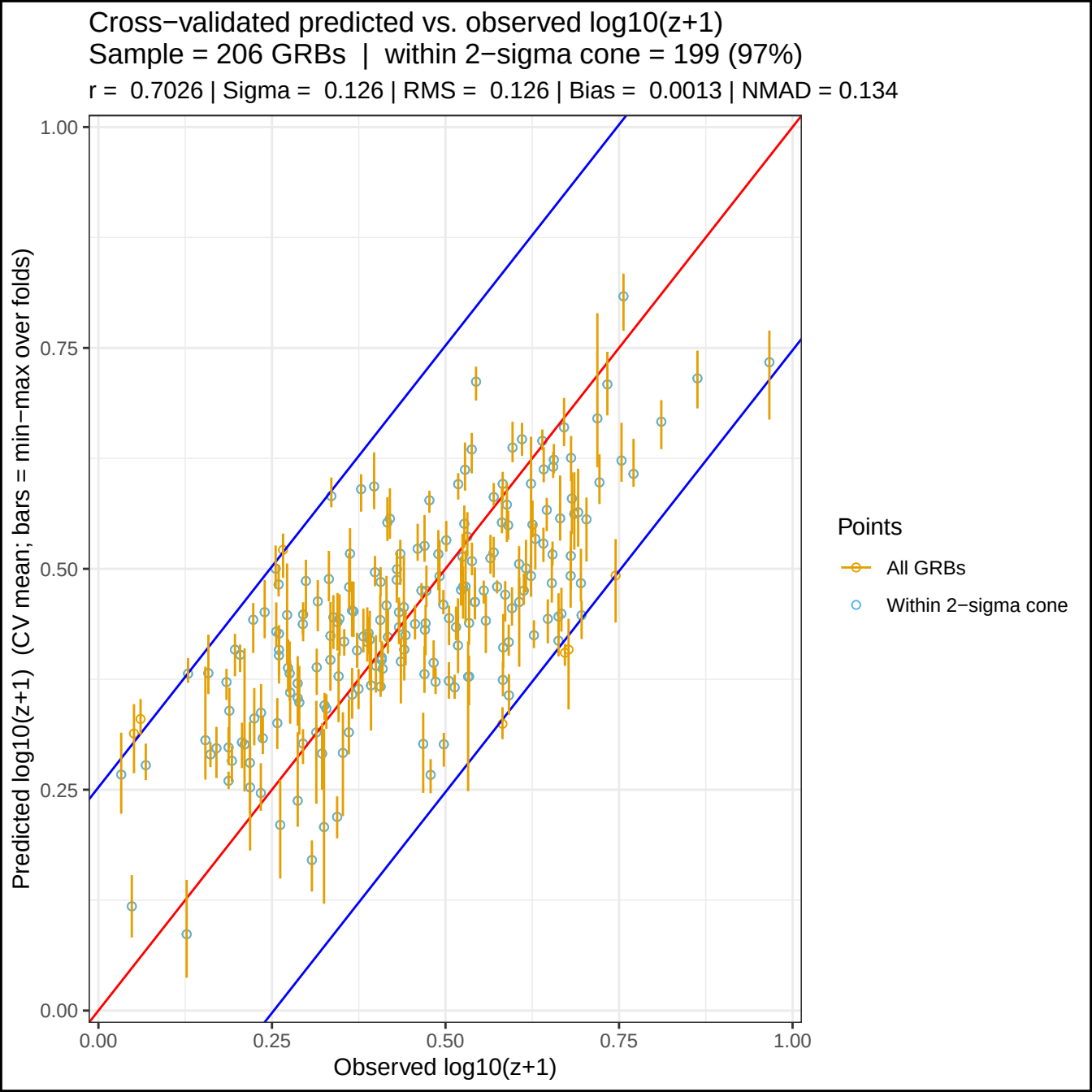

Redshift vs Bias
Bias
z
4
3
2
1
0
−1
0
2
4
6
8

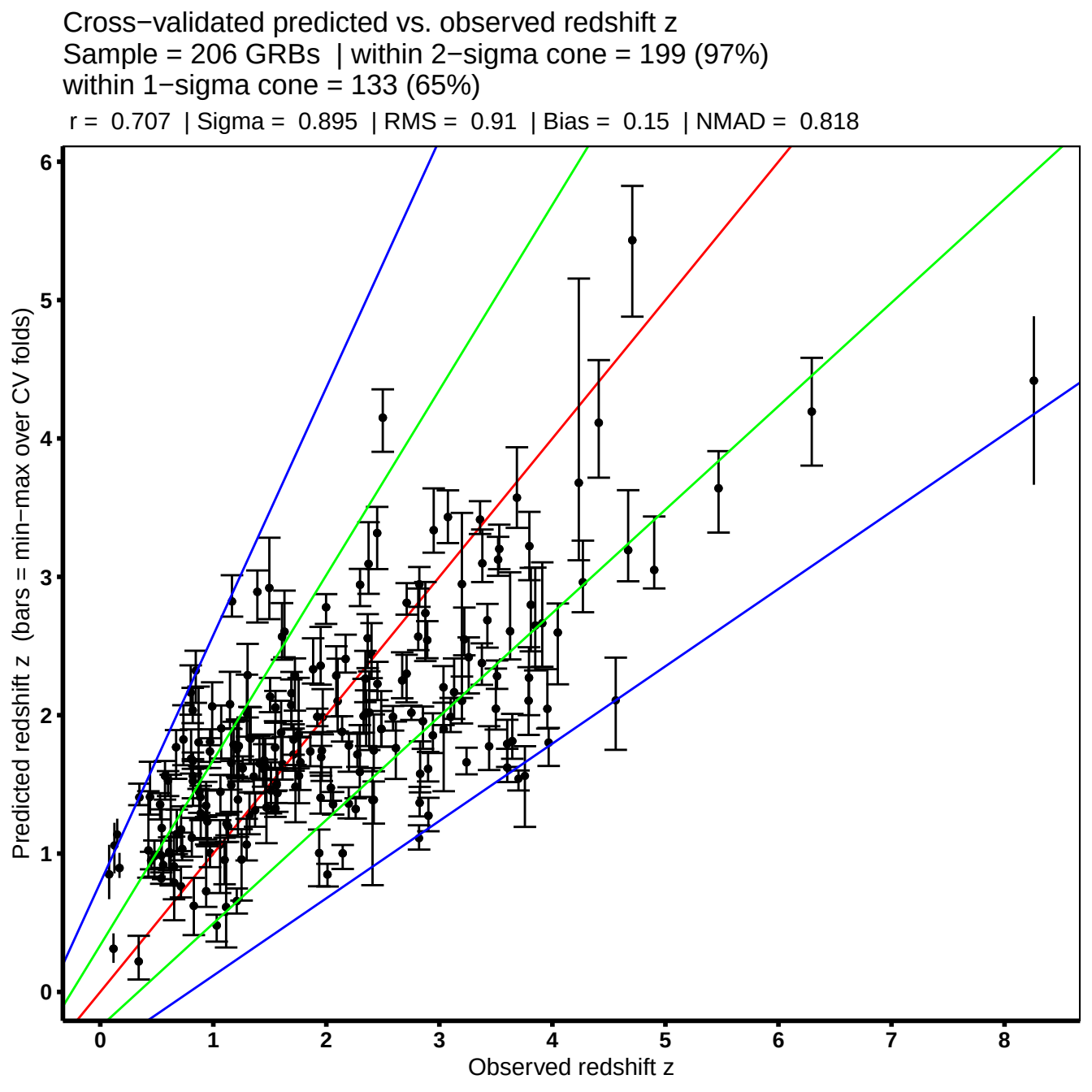

Redshift vs Normalized Bias
Normalized Bias
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
0
2
4
6
8
z

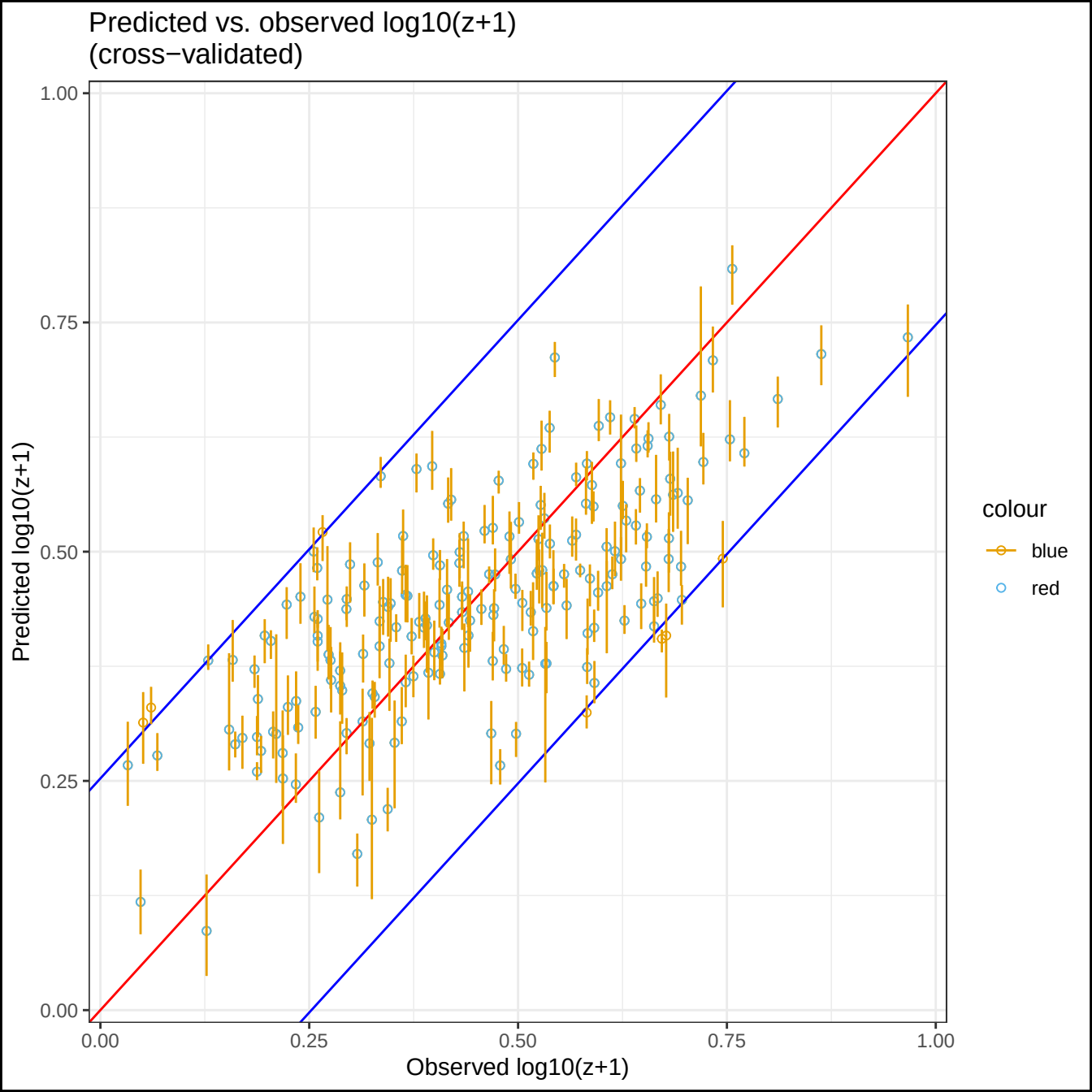

−0.2
−0.1
0.0
0.1
0.2

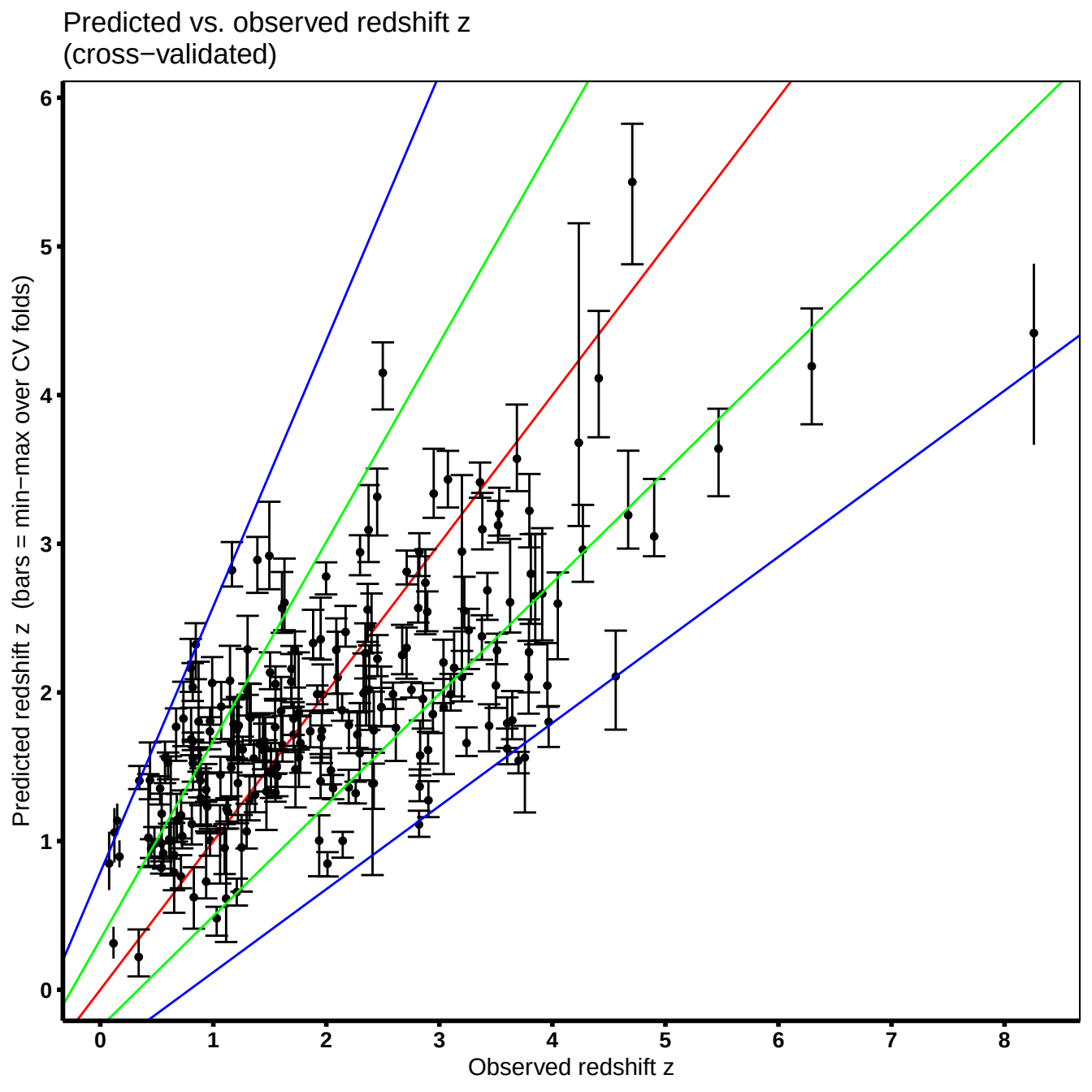

Redshift residual distribution (Dz = obs z − pred z)
Sigma=0.895, Bias=0.146
(blue = +/−Sigma, red = Bias)
Frequency
0
2
4
6
8
10
12
14
−1
0
1
2
3
4
Dz = observed z − predicted z


Normalized residual distribution (normalized Dz = (obs z − pred z)/(1+obs z))
Sigma=0.304, Bias=−0.0397
(blue = +/−Sigma, red = Bias)
Frequency
0
5
10
15
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
normalized Dz = (obs z − pred z)/(1 + obs z)


Cross−validated predicted vs. observed log10(z+1)
Sample = 206 GRBs | within 2−sigma cone = 199 (97%)
r = 0.7026 | Sigma = 0.126 | RMS = 0.126 | Bias = 0.0013 | NMAD = 0.134
Predicted log10(z+1) (CV mean; bars = min−max over folds)
Observed log10(z+1)
0.00
0.25
0.50
0.75
1.00
Points
All GRBs
Within 2−sigma cone


Cross−validated predicted vs. observed redshift z
Sample = 206 GRBs | within 2−sigma cone = 199 (97%)
within 1−sigma cone = 133 (65%)
r = 0.707 | Sigma = 0.895 | RMS = 0.91 | Bias = 0.15 | NMAD = 0.818
Predicted redshift z (bars = min−max over CV folds)
Observed redshift z
0
1
2
3
4
5
6
7
8


Predicted vs. observed log10(z+1)
(cross−validated)
Predicted log10(z+1)
Observed log10(z+1)
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
colour
blue
red


Predicted vs. observed redshift z
(cross−validated)
Predicted redshift z (bars = min−max over CV folds)
Observed redshift z
0
1
2
3
4
5
6
7
8